Christian DiPietrantonio
MES180: Dynamics
Project 02

Physical System

Body 01: Piston oriented at $+45°$ from horizontal (Track passes through origin)
- Can only translate (cannot rotate/move off track)

Body 02: Piston oriented at $-45°$ from horizontal (Track passes through origin)
- Can only translate (cannot rotate/move off track)

Body 03: Rigid Bar, $L = 0.10$ meters
- One end is connected to piston 1 via revolute/hinge joint
- Other end is connected to piston 2 via revolute/hinge joint
- Bar rotates at constant angular velocity, $\dot{\theta}_3 = \omega = 2$ rad/s

Generalized Coordinates

$$\vec{q} = \begin{bmatrix} x_1 \\ y_1 \\ \theta_1 \\ x_2 \\ y_2 \\ \theta_2 \\ x_3 \\ y_3 \\ \theta_3 \end{bmatrix}$$

x_1, y_1: Position of Piston 1 Center (Global Frame)
θ_1 → Orientation of Piston 1
x_2, y_2: Position of Piston 2 Center (Global Frame)
θ_2 → Orientation of Piston 2
x_3, y_3: Position of Rigid bar center (Global Frame)
θ_3 → Orientation of Rigid bar

Rotation Matrix:

* Inverse of $A(\theta)$ = transpose

$$\left.\begin{array}{l} R^{-1} \cdot R = I \\ R^T \cdot R = I \end{array}\right\} R^{-1} = R^T$$

Harder to calculate

$$A(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

* Trig Identities

$\cos(-\theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$

$$A(\theta)^T = A(-\theta)^T$$

$$\begin{bmatrix} V_x \\ V_y \end{bmatrix}_{Global} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} V'_x \\ V'_y \end{bmatrix}_{local}$$

$$\begin{bmatrix} V_x \\ V_y \end{bmatrix}_{Global} \underbrace{\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}}_{A(\theta)^T} = \begin{bmatrix} V'_x \\ V'_y \end{bmatrix}_{local}$$

Locating Points on Rigid Bodies

If body i has center at $\vec{R}_i = [x_i, y_i]^T$ and orientation θ_i, then Point P located at $\vec{s}^{(i)} = [S_x, S_y]^T$ in the body's local frame has global position:

$$\vec{r}_P = \vec{R}_i + A(\theta_i)\vec{s}^{(i)}$$

$$\begin{bmatrix} r_{P,x} \\ r_{P,y} \end{bmatrix} = \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{bmatrix} \begin{bmatrix} S_x \\ S_y \end{bmatrix}$$

$$= \begin{bmatrix} x_i + S_x\cos\theta_i - S_y\sin\theta_i \\ y_i + S_x\sin\theta_i + S_y\cos\theta_i \end{bmatrix}$$

For our System

- Pistons 1 & 2 (hinges are mounted at center of each piston, so hinge in each pistons local frame is at the origin: $\vec{s}_1 = \vec{s}_2 = [0,0]^T$

$$\vec{r}_{hinge,1} = \vec{R}_1 + A(\theta_1)\vec{s}_1$$

$$= \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$\vec{r}_{hinge,2} = \vec{R}_2 + A(\theta_2)\vec{s}_2$$

$$= \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

- Rigid Bar (Bar has length L & local x-axis runs along its length. Center at $\vec{R}_3 = [x_3, y_3]^T$
  - End A: Connects to piston 1 at local coordinates $\vec{S}_{3A} = [-L/2, 0]^T$

    In global coordinates:

$$\vec{r}_A = \vec{R}_3 + A(\theta_3)\vec{S}_{3A}$$

$$= \begin{bmatrix} x_3 \\ y_3 \end{bmatrix} + \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 \\ \sin\theta_3 & \cos\theta_3 \end{bmatrix} \begin{bmatrix} -L/2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} x_3 - L/2\cos\theta_3 \\ y_3 - L/2\sin\theta_3 \end{bmatrix}$$

• End B: Connects to piston 2 at local coordinates $\vec{s}_{3B} = [L/2, 0]^T$

$$\vec{r}_B = \vec{R}_3 + A(\theta_3)\vec{s}_{3B}$$

$$= \begin{bmatrix} x_3 \\ y_3 \end{bmatrix} + \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 \\ \sin\theta_3 & \cos\theta_3 \end{bmatrix} \begin{bmatrix} L/2 \\ 0 \end{bmatrix}$$

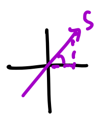
$$= \begin{bmatrix} x_3 + L/2 \cos\theta_3 \\ y_3 + L/2 \sin\theta_3 \end{bmatrix}$$

Building Constraint Equations

• We need 9 constraint equations for 9 DOF

Contsraint Type 1: Prismatic (Sliding) Joint - Piston 1 on the + 45° Track

• Constraint C_1: Piston 1 position on track



$$\left.\begin{matrix} x = s\cos(45) \\ y = s\sin(45) \end{matrix}\right\} \begin{bmatrix} x \\ y \end{bmatrix} = s \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \rightarrow \left.\begin{matrix} x = s/\sqrt{2} \\ y = s/\sqrt{2} \end{matrix}\right\} \underline{y = x}$$

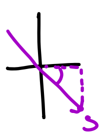
For center of piston 1: $\boxed{C_1 = y_1 - x_1 = 0}$

• Constraint C_2: Piston 1 orientation:

$$\boxed{C_2 = \theta_1 - \pi/4 = 0}$$

Contsraint Type 2: Prismatic (Sliding) Joint - Piston 2 on the −45° Track

• Constraint C_3: Piston position on track



$$\left.\begin{matrix} x = s\cos(45) \\ y = s\sin(45) \end{matrix}\right\} \begin{bmatrix} x \\ y \end{bmatrix} = s \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \rightarrow \left.\begin{matrix} x = s/\sqrt{2} \\ y = -s/\sqrt{2} \end{matrix}\right\} \underline{y = -x}$$

For center of piston : $\boxed{C_3 = y_2 + x_2 = 0}$

• Constraint C_4: Piston orientation:

$$\boxed{C_4 = \theta_2 + \pi/4 = 0}$$

Contsraint Type 3: Revolute (Hinge) Joint A - Piston 1 to Bar end A

- Revolute joint forces two points (one on each body) to coincide in space. (gives 2 Scaler con
- The hinge connects the center of Piston 1 to end A of the Bar. These two points must have the same global position

$$\vec{r}_{hinge,1} = \vec{r}_A$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_3 - L/2 \cos\theta_3 \\ y_3 - L/2 \sin\theta_3 \end{bmatrix}$$

X-Component:

$$\boxed{C_5 = x_1 - x_3 + L/2 \cos\theta_3 = 0}$$

Y-component:

$$\boxed{C_6 = y_1 - y_3 + L/2 \sin\theta_3 = 0}$$

Contsraint Type 4: Revolute (Hinge) Joint B - Piston 2 to Bar end B

$$\vec{r}_{hinge,2} = \vec{r}_B$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_3 + L/2 \cos\theta_3 \\ y_3 + L/2 \sin\theta_3 \end{bmatrix}$$

X-Component:

$$\boxed{C_7 = x_2 - x_3 - L/2 \cos\theta_3 = 0}$$

Y-Component:

$$\boxed{C_8 = y_2 - y_3 - L/2 \sin\theta_3 = 0}$$

Contsraint Type 5: Driving Constraint

- The bar rotates at a constant pescribed rate $\dot{\theta}_3 = \omega = 2$ rad/s. Integrating this with $\theta_3(0) = 0$:

$$\theta_3 = \omega t$$

$$\boxed{C_9 = \theta_3 - \omega t = 0}$$

Assembling the constraint vector:

- We have 9 equations and 9 unknowns, System is fully determined at each time t.

$$\vec{C}(\vec{q},t) = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \\ C_5 \\ C_6 \\ C_7 \\ C_8 \\ C_9 \end{bmatrix} = \begin{bmatrix} y_1 - x_1 \\ \theta_1 - \pi/4 \\ y_2 + x_2 \\ \theta_2 + \pi/4 \\ x_1 - x_3 + L/2\cos\theta_3 \\ y_1 - y_3 + L/2\sin\theta_3 \\ x_2 - x_3 - L/2\cos\theta_3 \\ y_2 - y_3 - L/2\sin\theta_3 \\ \theta_3 - \omega t \end{bmatrix} = \vec{0}$$

The Constraint Jacobian

$$C_q = \frac{\partial \vec{C}}{\partial \vec{q}}$$

* len($\vec{q}$) columns
len($\vec{c}$) rows

$$C_q = \begin{bmatrix} \frac{\partial C_1}{\partial x_1} & \frac{\partial C_1}{\partial y_1} & \cdots & \frac{\partial C_1}{\partial \theta_3} \\ \frac{\partial C_2}{\partial x_1} & \frac{\partial C_2}{\partial y_1} & \cdots & \frac{\partial C_2}{\partial \theta_3} \\ \vdots & & \ddots & \\ \frac{\partial C_9}{\partial x_1} & \frac{\partial C_9}{\partial y_1} & \cdots & \frac{\partial C_9}{\partial \theta_3} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -L/2\sin\theta_3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & L/2\cos\theta_3 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & L/2\sin\theta_3 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & -L/2\cos\theta_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The Partial Time derivative vector

$$\vec{C}_t = \frac{\partial \vec{C}}{\partial t}$$

$$\vec{C}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\omega \end{bmatrix}$$

The velocity equation

*MVCR:

$$\frac{d\vec{C}}{dt} = \sum_i \frac{\partial \vec{C}}{\partial x_i}\frac{dx_i}{dt} + \frac{\partial \vec{C}}{\partial t}$$

$$\frac{d}{dt}\vec{C} = \frac{\partial \vec{C}}{\partial q} \cdot \frac{d\vec{q}}{dt} + \frac{\partial \vec{C}}{\partial t} = \vec{0}$$

$$C_q \dot{\vec{q}} + \vec{C}_t = \vec{0}$$

$$\boxed{C_q \dot{\vec{q}} = -\vec{C}_t}$$

$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & -L/2 \sin\theta_3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & L/2 \cos\theta_3 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & L/2 \sin\theta_3 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & -L/2 \cos\theta_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{\theta}_1 \\ \dot{x}_2 \\ \dot{y}_2 \\ \dot{\theta}_2 \\ \dot{x}_3 \\ \dot{y}_3 \\ \dot{\theta}_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega \end{bmatrix}$$

The Acceleration Equation

- Starting with the velocity equation:

$$C_q \dot{\vec{q}} + \vec{C}_t = \vec{0}$$

- Taking total time derivative

$$\frac{d}{dt}\left(C_q \dot{\vec{q}}\right) + \frac{d}{dt}\left(\vec{C}_t\right) = \vec{0}$$

$$\downarrow$$

$$\frac{dC_q}{dt}\dot{\vec{q}} + C_q \ddot{\vec{q}}$$

$$C_q \ddot{\vec{q}} + \frac{dC_q}{dt}\dot{\vec{q}} + \frac{d\vec{C}_t}{dt} = \vec{0}$$

$$\boxed{C_q \ddot{\vec{q}} = \underbrace{-\frac{dC_q}{dt}\dot{\vec{q}} - \frac{d\vec{C}_t}{dt}}_{\vec{\gamma}}}$$

Computing the gamma vector

Term 2:
$$\frac{dC_t}{dt} = \vec{0}$$

Term 1: Since Θ_3 changes with time, $\frac{d}{dt} f(\Theta_3) = \frac{df}{d\Theta_3}\dot{\Theta}_3$

$$-L/2 \sin\Theta_3 \longrightarrow -L/2 \cos\Theta_3 \cdot \dot{\Theta}_3$$

$$L/2 \cos\Theta_3 \longrightarrow -L/2 \sin\Theta_3 \cdot \dot{\Theta}_3$$

$$L/2 \sin\Theta_3 \longrightarrow L/2 \cos\Theta_3 \cdot \dot{\Theta}_3$$

$$-L/2 \cos\Theta_3 \longrightarrow L/2 \sin\Theta_3 \cdot \dot{\Theta}_3$$

$$\frac{dC_q}{dt} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -L/2\cos\theta_3 \cdot \dot{\theta}_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -L/2\sin\theta_3 \cdot \dot{\theta}_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & L/2\cos\theta_3 \cdot \dot{\theta}_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & L/2\sin\theta_3 \cdot \dot{\theta}_3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

for $\frac{dC_q}{dt}\dot{\vec{q}}$, only column 9 contributes and it gets multiplied by $\dot{\vec{q}} = \dot{\theta}_3$

$$\frac{dC_q}{dt}\dot{\vec{q}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -L/2\cos\theta_3 \cdot \dot{\theta}_3^2 \\ -L/2\sin\theta_3 \cdot \dot{\theta}_3^2 \\ L/2\cos\theta_3 \cdot \dot{\theta}_3^2 \\ L/2\sin\theta_3 \cdot \dot{\theta}_3^2 \\ 0 \end{bmatrix}$$

$$\vec{\gamma} = -\frac{dC_q}{dt}\dot{\vec{q}} - \frac{d\vec{C}_t}{dt}$$

$$= -\frac{dC_q}{dt}\dot{\vec{q}} - \vec{0} \rightarrow \boxed{\vec{\gamma} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ L/2\cos\theta_3 \cdot \dot{\theta}_3^2 \\ L/2\sin\theta_3 \cdot \dot{\theta}_3^2 \\ -L/2\cos\theta_3 \cdot \dot{\theta}_3^2 \\ -L/2\sin\theta_3 \cdot \dot{\theta}_3^2 \\ 0 \end{bmatrix}}$$

Solution Process

- Perform 3 sequential solves at each time step t_i:
- Step 1: Position Analysis (Non-Linear Solve) /Transcendental

↳ find $\vec{q}$ such that $\vec{C}(\vec{q}, t_i) = 0$

↳ Non-Linear → use Newton Raphson iterative solver

↳ use solution from previous step as initial guess

- Step 2: Velocity Analysis (Linear Solve)

↳ Solve $C_q \dot{\vec{q}} = -\vec{C}_t$ for $\dot{\vec{q}}$

↳ LU Factorization (\ Julia operator)

↳ Need position $\vec{q}$ from Step 1 to get C_q and $\vec{C}_t$

- Step 3: Acceleration Analysis (Linear Solve)

↳ Solve $C_q \ddot{\vec{q}} = \vec{\gamma}$ for $\ddot{\vec{q}}$

↳ LU Factorization

↳ Both $\vec{q}$ from Step 1 and $\dot{\vec{q}}$ from Step 2 are needed to evaluate $\vec{\gamma}$

Algorithm Summary

for each time step t_i:

1. Solve $\vec{C}(\vec{q}, t_i) = 0$ → get $\vec{q}$ (Positions)
2. Solve $C_q \dot{\vec{q}} = -\vec{C}_t$ → get $\dot{\vec{q}}$ (velocities)
3. Solve $C_q \ddot{\vec{q}} = \vec{\gamma}$ → get $\ddot{\vec{q}}$ (accelerations)
4. Store $\vec{q}, \dot{\vec{q}}, \ddot{\vec{q}}$
5. Update initial guess for next step

TO DO:

- Code
- Results
- Conclusions
- Reproducing Results ⤳ * Check to see if we have used all modules before *
- Project Dependencies
- Finish Project Structure
- Analytical Solution Validation (IF TIME)

Analytical Sol

$$\frac{L}{2}\left[\sin\theta_3 - \cos\theta_3 - \sin\theta_3 - \cos\theta_3\right]$$

$$2y_3 = \frac{L}{2}\left[-2\cos\theta_3\right]$$

$$y_3 = -\frac{L}{2}\cos\theta_3$$

$$2x_3 = -\frac{L}{2}\left[\sin\theta_3 + \cos\theta_3 + \sin\theta_3 - \cos\theta_3\right]$$

$$= -\frac{L}{2}\, 2\sin\theta_3$$

$$x_3 = -\frac{L}{2}\sin\theta_3$$

$$x_1 = -\frac{L}{2}\overset{\omega\cos\theta_3}{\sin\theta_3} - \frac{L}{2}\overset{-\omega\sin\theta_3}{\cos\theta_3}$$

$$\dot{x}. = -\frac{L}{2}\cos\theta_3\dot{\theta}_3 + \frac{L}{2}\sin\theta_3\dot{\theta}_3$$

$$= \frac{L}{2}\left[\sin\theta_3 - \cos\theta_3\right]\dot{\theta}_3$$

$$x_2 = \frac{L}{2}\cos\theta_3 - \frac{L}{2}\sin\theta_3$$

$$-\frac{L}{2}\sin\theta_3 - \frac{L}{2}\cos\theta_3$$

$$-\frac{L}{2}\left[\sin\theta_3 + \cos\theta_3\right]\omega$$

$$x_3 = -\frac{L}{2}\sin\theta_3 \qquad y_3 = -\frac{L}{2}\cos\theta_3$$

$$\dot{x}_3 = -\frac{L}{2}\cos\theta_3\,\omega \qquad \dot{y}_3 = \frac{L}{2}\sin\theta_4\,\omega$$

$$\dot{x}_1 = \left(\frac{L}{2}\sin\theta_3 - \frac{L}{2}\cos\theta_3\right)\omega$$

$$\ddot{x}_1 = \frac{L}{2}\cos\theta_3 + \frac{L}{2}\sin\theta_3 \;\omega^2$$

$$\dot{x}_2 = \left(-\frac{L}{2}\sin\theta_3 \;\; -\frac{L}{2}\cos\theta_3\right)\omega$$

$$\ddot{x}_2 = -\frac{L}{2}\cos\theta_3 + \frac{L}{2}\sin\theta_3 \;\; \omega^2$$

$$\dot{x}_3 = -\frac{L}{2}\cos\theta_3\,\omega \qquad \dot{y}_3 = \frac{L}{2}\sin\theta_3\,\omega$$

$$\ddot{x} = \frac{L}{2}\sin\theta_3\,\omega^2 \qquad \ddot{y}_3 = \frac{L}{2}\cos\theta_3\,\omega^2$$

$\omega = 2$ rad/s $\qquad$ 1 full rotation $= 2\pi$ rad

$$\theta(t) = \omega t$$

$$\theta(t_{end}) = 2\pi$$

$$2\pi = \omega \cdot t_{end} \rightarrow \boxed{t_{end} = \pi \text{ seconds}}$$